



Electronic nose system for real-time burnt odor detection and multi-stage cooking recognition in smart ovens

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ABSTRACT

Burnt odors, volatile organic compounds (VOCs) released by overcooked food, serve as early indicators of kitchen fire hazards. However, existing gas sensing technologies lack the capability to detect burnt odors across diverse cooking scenarios. We present an electronic nose (E-nose) system integrated with a household oven for multi-stage cooking recognition. The E-nose comprises four metal-oxide semiconductor (MOS) gas sensors coupled with a water-cooling gas-path structure designed to condition the humidity and oil contamination of oven gas. Four representative recipes covering common food categories were selected to evaluate system performance. The proposed system continuously monitors the entire roasting process, which is further defined into four stages: raw, tender, crispy, and burnt. By employing a window-based feature extraction strategy combined with the Extreme Gradient Boosting (XGBoost) algorithm, the system achieved approximately 90% classification accuracy in both single-device and cross-device offline evaluations. Furthermore, the trained model was deployed on an embedded board within a new E-nose device for real-time inference across 20 intensive cooking trials, yielding an average accuracy of 91.3% and demonstrating initial stability of the system. Additionally, unseen-recipe validation complemented by trend analysis on six additional recipes provided preliminary evidence of the system's generalizability to novel ingredients. This work offers a promising technology for applying electronic nose in smart kitchen safety monitoring and intelligent home appliances.

1. Introduction

Unattended cooking equipment is identified by the National Fire Protection Association (NFPA) as the leading cause of residential fires and casualties [1–3]. Its prolonged heating leads to the carbonization of food, releasing pungent volatile organic compounds (VOCs) that serve as precursor warning signals prior to visible combustion. Beyond fire risks, excessive browning reactions generate hazardous byproducts [4–7], including acrylamide, hydroxymethylfurfural, and heterocyclic amines [4,5]. These compounds are associated with neurotoxicity, carcinogenicity, and other adverse health effects [8,9]. Therefore, the real-time monitoring of cooking doneness and the early detection of burnt odors are critical for ensuring both user safety and food health.

Electronic noses (E-noses) have demonstrated significant efficacy in analyzing complex gas mixtures for environmental monitoring [10–13] and hazardous gas detection [14–16]. In culinary applications, E-nose arrays have been successfully employed to characterize cooking activities [17–19], ingredients [20], and air quality [21,22]. Recent studies have further explored the detection of specific cooking states, such as the frying status of foods [23], the roasting stages of chicken via sensor fusion [24], and the baking status of bread using commercial gas sensors [25–27].

Despite these advancements, current research predominantly focuses on unburnt stages [28–31] or relies on single-recipe datasets [24–27,32,33], with limited investigation into the complex odor profiles

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generated during severe charring across diverse food types (see Supplementary Materials Section 1). Furthermore, practical deployment in kitchen environments faces substantial engineering hurdles. E-nose systems must maintain sensor activity and stability while withstanding harsh conditions, including high temperatures, high humidity, and severe oil contamination [34–36]. Achieving real-time response, multi-recipe adaptability, and cross-device compatibility remains a critical challenge for system integration [37,38].

In this work, we present a smart E-nose system designed for real-time burnt odor detection and multi-stage cooking monitoring, using electric ovens as a representative case study. The system incorporates a custom array of four self-developed metal oxide semiconductor (MOS) sensors integrated with a compact water-cooling gas path. This structural design enables the system to effectively sample gases from high-temperature and oily oven environments. To validate the system, we defined four fine-grained cooking stages (raw, tender, crispy, and burnt) across four distinct food categories: toast, egg tarts, chicken wings, and sausages. The “burnt” stage is defined as the point of significant carbonization, representing the most severe overcooking condition.

To elucidate the sensing mechanism, we characterized the VOC profiles of these cooking stages using gas chromatography–mass spectrometry (GC–MS). Representative odorant markers were identified, and static gas distribution tests were conducted to verify sensor response. For data processing, we implemented a window-based sampling strategy coupled with an Extreme Gradient Boosting (XGBoost) classifier [39]. The system achieved an approximately 90% accuracy in both single-device and cross-device offline evaluations. Furthermore, the model was deployed on an embedded board within a new E-nose device for online inference across 20 intensive cooking trials, yielding an average accuracy of 91.3% and demonstrating initial stability of the system. To explore the system’s generalizability on new recipes, we implemented a unified model trained on all four recipes, and a “leave-one-recipe-out” validation to simulate the processing of unseen recipes. Additionally, data of six new recipes were also analyzed to investigate our system’s broader applicability. This study provides a promising gas sensing solution for various oven-cooking recipes, advancing the practical application of E-nose technology in smart kitchen ecosystems.

2. Hardware design and system implementation

2.1. Electronic nose design

The E-nose system comprises four primary modules: a custom sensor array, control circuitry, a water-cooling gas chamber, and dual micro-pumps, as illustrated in Fig. 1(a).

- **Sensor array.** The sensor array incorporates four self-developed MOS sensors (S1–S4), rather than commercial alternatives. SnO_2 and WO_3 were synthesized as the primary sensing materials owing to their broad-spectrum sensitivity toward VOCs (shown in Table 1). The comprehensive fabrication protocols and gas-sensing performance evaluations of these specific sensors have been detailed in our previous work [40]. To ensure environmental robustness, a commercial temperature and humidity sensor (Sensirion SHT40) was integrated at the center of the array for real-time compensation.
- **Circuitry.** The hardware utilizes a dual-board architecture. The upper board integrates the sensor array, sensor power supplies, and signal outputs and the lower board contains a microcontroller unit (MCU), power supplies, voltage regulators, pump control modules, and other necessary components (Fig. 1(b–c)). Sensor responses are measured via a voltage divider circuit (Fig. 1(d) and Formula (1)), where the voltage across the load resistor (R_L) is digitized by the MCU at a sampling rate of 10 Hz.

$$\text{Response} = V_{R_L} = \frac{R_L}{R_L + R_{\text{sensor}}} * V_{cc} \quad (1)$$

Table 1

Materials of 4 self-developed MOS sensors [40].

Label	Sensor manufacturer	Material
S1	The Future Laboratory, Tsinghua University	Pd/SnO_2
S2	The Future Laboratory, Tsinghua University	Pd/WO_3
S3	The Future Laboratory, Tsinghua University	Pt/SnO_2
S4	The Future Laboratory, Tsinghua University	Pt/WO_3

- **Gas path and water-cooling structure.** The gas atmosphere inside ovens is featured with up to 230 °C, 100% RH, and severe pollution of oil and smoke during cooking. In order to reduce the humidity and pollution of extracted air, a monolithic 3D-printed structure combining a gas chamber and a water-cooling structure was designed. The water-cooling structure features concentric channels that guide airflow through a coolant interface before reaching the sensors (seen in Fig. 1(e) and (f)). The sensor chamber is secured to the circuit board with eight screws and equipped with an O-ring seal to enhance the airtightness.

Two diaphragm pumps (6 V, 1.1 L/min) regulate the airflow, enabling alternating cycles of sampling and purging to clean the sensor chamber. For cleaning process, the sweeping pump drives the outside air to the sensor chamber and the water-cooling paths. (Fig. 1(e)).

2.2. Integration with oven

A commercial oven was chosen for the experiment. The E-nose was connected to the oven chamber via a steel pipe. One end of the steel pipe is connected to the electronic nose gas inlet, and the other end is connected to the hole drilled on the oven inner chamber which is located on the top of the side wall. Fig. 1(g) and (h) shows the integration of our E-nose and the oven.

2.3. Software and sampling strategy

The E-nose device runs CircuitPython² environment for sensor data reading, signal communication, and model inference. During the dataset collection experiments, the E-nose runs as a slave device and a PC running python environment was used as data recorder. During the real-time test, no PC was used as all feature extraction and cooking stage recognition were conducted on board.

Considering that the oven baking process is a slow-changing process, we designed a window-based sampling strategy by simulating the respiratory mechanism of biological sniffing processes. This alternating sampling method of inhale–exhale cycles also helps prolong the lifespan of the E-nose. A data window consists of a 4-second sampling period and a 116-second sweeping period, with 2 min as one cycle. The sensor values increase during the sampling period and reach their peaks within 2 s after the sweeping pump begins to work due to the delay in the gas path, then decrease and return to the baseline after sweeping. The above sampling cycles are repeated throughout the baking process. Fig. 2(a) shows the odor data of a whole baking process (S1, sausage), and Fig. 2(b) shows the first 30 s of a single data window.

3. Experiment design

3.1. Oven recipe selection

In this study, four common recipes were selected as experimental cases. Due to the fact that cakes&pastries and meats are the two main categories for oven recipes, we chose toasts, egg tarts, sausages,

² CircuitPython: <https://circuitpython.org/>.

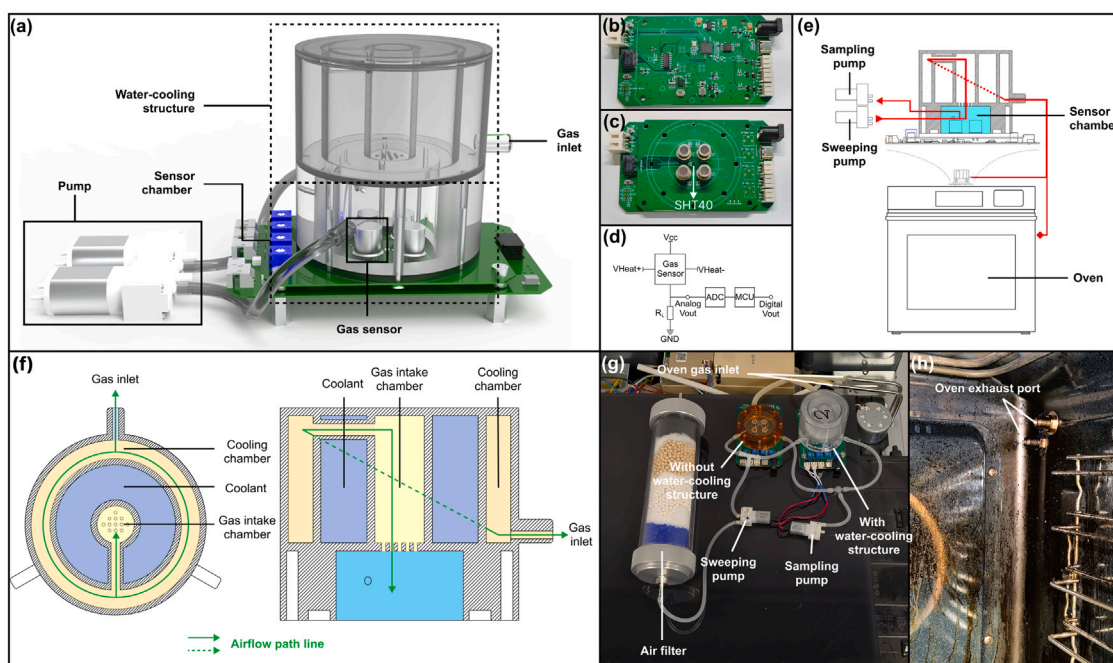


Fig. 1. (a) The E-nose device comprises four gas sensors. (b–c) E-nose circuit board. (d) Functional block diagram of the circuit. (e) Schematic diagram of the gas path in the e-nose system. (f) Water-cooling structure and airflow path diagram (cross-section top view and main view). (g) Experimental setup for water-cooling structure evaluation. (h) Two parallel sampling holes drilled on the side wall of the oven.

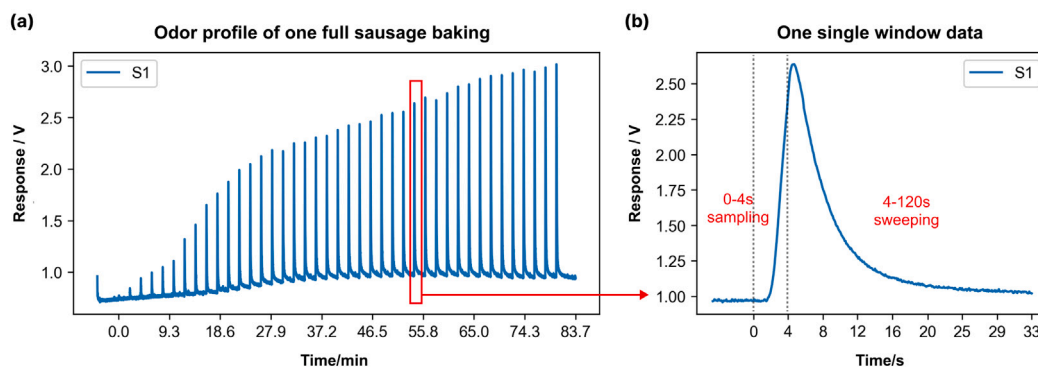


Fig. 2. Visualization of the E-nose sensor data. (a) illustrates the dynamic response profile of S1 during a full sausage baking cycle, spanning from beginning (0 min) to complete burnt (≈ 80 min). The window-based sampling strategy generates characteristic periodic spike sequence patterns. (b) provides a zoomed-in view of a single data window, distinctly showing the sampling phase (0–4 s), where the peak occurs within 2 s after sweeping initiation (attributable to pneumatic latency in gas path configuration).

and chicken wings as four testing recipes. For detecting the normal cooking state as well as the burnt states, we intentionally set the cooking time longer than normal conditions in order to fully burn the foods and collected odor data throughout the whole process. Detailed methods for preparing recipes and ingredient information are reported in Supplementary Materials Section 2.

3.2. Human evaluation for cooking stage definition

To establish ground truth labels for supervised learning, we designed a human evaluation experiment. Seven food assessors were trained on standardized definitions derived from culinary consensus and determined the four cooking time points of each recipe based on the baking videos. These time points demarcated four distinct cooking stages, serving as the classification targets for the E-nose system, as detailed in Table 2. Each recipe was baked for 10 times, during which 40 videos of food baking process were captured. Detailed averaging calculation was presented in the result section.

3.3. GC–MS analysis and sensor applicability evaluation

To characterize the VOC profiles evolved during roasting, GC–MS was employed to analyze odor compounds across the cooking stages defined in Table 2. (Detailed methodologies are provided in Supplementary Materials Section 3).

To validate the sensor array's response to characteristic cooking VOCs, target analytes were selected by correlating our GC–MS results with established literature on food volatiles. For each recipe, two representative compounds were chosen: a “global” marker present throughout the entire cooking process (stages t0–t3) and a “burnt” marker specific to the overcooking phase (stages t2–t3). In total, eight compounds were evaluated at concentrations of 1, 5, and 10 ppm to assess sensor linearity and sensitivity across relevant ranges.

Experiments were conducted using a static gas distribution method within a fume hood, as illustrated in Fig. 3. The E-nose was enclosed in a testing chamber where liquid analytes were injected onto a heating plate and vaporized. Internal fans were utilized to homogenize the

Table 2
Cooking stage definition.

Cooking point	Description	Cooking stage	Definition
t0	The food begins to ripen.	0–t0	Raw stage, inedible
t1	The food starts to be crispy and is slightly overcooked. You still being willing to eat.	t0–t1	Tender stage, edible
t2	The food begins to be burnt and unhealthy to eat. You do not want to eat in normal situations, unless you are extremely hungry.	t1–t2	Crispy stage, edible
t3	The food has no edible value and can be discarded. You do not want to eat under any conditions.	t2–t3	Burnt stage, inedible

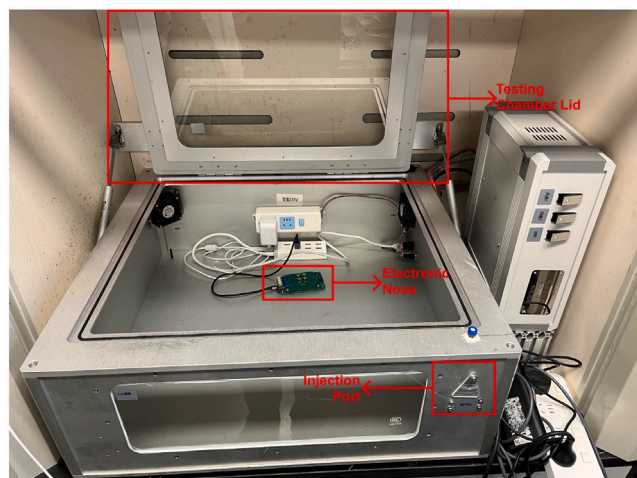


Fig. 3. Schematic of the static gas distribution device employed for sensor testing.

gas mixture. Sensor responses were recorded until a steady state was achieved (approximately 5 min), followed by a ventilation phase to restore baseline levels before subsequent testing cycles.

3.4. Water-cooling structure evaluation

To evaluate the thermal and humidity management efficacy of the water-cooling structure, a comparative experimental setup was established. Two E-nose systems were developed, sharing identical circuitry, sensors, and micro-pumps. The experimental group utilized the proposed water-cooling structure, while the control group employed a standard chamber. To ensure structural integrity under high-temperature stress and prevent potential pneumatic leakage, the control chamber was fabricated from heat-resistant photosensitive resin. In contrast, the water-cooling version was produced using transparent photosensitive resin, as shown in Fig. 1(g).

To minimize environmental interference and ensure consistent baseline air quality, the intake ports of both systems' sweeping pumps were connected in parallel to a multi-stage air filter. This filtration unit contained activated carbon, molecular sieves, activated alumina, and allochroic silica gel, arranged sequentially to strip ambient VOCs and moisture. During testing, both E-noses were mounted on a single oven with sampling ports drilled 2.5 cm apart on the oven wall (Fig. 1(h)). This dual-sampling configuration ensures that any observed differences in internal humidity and temperature are attributable to the structural design rather than variations in oven conditions or ingredient composition.

3.5. Cooking experiments

A series of experiments were designed to collect cooking odor profiles and evaluate the performance of the classification algorithm. Initially, two ovens and three E-nose devices (#1–#3) were employed to construct a comprehensive odor dataset across four recipes, which

Table 3
The ground-truth cooking points for each recipe.

Recipe	t0/min	t1/min	t2/min	t3/min
Toast	10	13	17	29
Egg tart	17	25	29	42
Chicken wing	17	26	35	57
Sausage	15	29	46	79

served as the foundation for model development as well as intra- and cross-device evaluation. To rigorously evaluate the system's robustness and real-world compatibility, a fourth E-nose (#4) was utilized for real-time validation with the trained model embedded on-board. This real-time evaluation was conducted in a distinct geographical location (Foshan, South China) characterized by higher ambient temperatures and humidities than the initial experimental site (Beijing). Each session was maintained until severe food charring occurred, thereby subjecting the sensor array to extreme conditions to test its long-term durability and stability.

It should be emphasized that environmental parameters, including ambient temperature, humidity, and airflow, were not externally regulated but maintained at natural room conditions. This experimental design was intended to simulate the typical operational environment of household ovens, thereby evaluating the E-nose's performance and robustness in practical, real-world applications rather than under idealized, strictly controlled laboratory settings.

4. Results and discussion

4.1. The distinction of cooking stages

We calculated the average selected time for each recipe, first averaging 10 cooking sessions, then averaging participants. The highest- and lowest-average participants were discarded, remaining five assessors for the final mean values. Quantitative analysis of inter-rater consistency has been conducted (See Supplementary Materials Section 4). Table 3 shows the final cooking points. These results are regarded as the ground-truth cooking stages and used for labeling the odor data. It is worth noting that the determination of cooking time points in this paper is to validate the multi-stage cooking recognition ability of the E-nose system and is not the core contribution. The system and algorithm proposed in this paper can also be adapted to other time point divisions.

4.2. Analysis of VOCs at different cooking stages

The GC–MS tests were performed according to the time points specified in Table 3. A total of 25 compounds were retained for toast, 24 for sausages, 32 for chicken wings, and 26 for egg tarts. The detailed selection rules and results of the retained substances are presented in Supplementary Materials Section 6.

To evaluate the response performance of the self-developed sensors in the oven task, eight representative odorant compounds were selected (see Table 4). The selected compounds were detected in the GC–MS results of this work with high similarity and reproducibility, while also reported in previous related research, verifying the credibility of the present results. A series of static gas testing experiments were

Table 4
The selected odorant compounds for sensor testing.

Recipe	Stage	Odorant	CAS	Purity	Reference
Toast	All stages	N-hexanal	66-25-1	99.5%	[41–43]
	Burnt	5-hydroxymethylfurfural	67-47-0	99.0%	[44,45]
Egg tart	All stages	Furfural	98-01-1	99.0%	[41–43]
	Burnt	Valeraldehyde	110-62-3	98.0%	[46]
Chicken wing	All stages	1-nonanal	124-19-6	98.0%	[47,48]
	Burnt	Acrolein	107-02-8	95.0%	[49]
Sausage	All stages	Heptaldehyde	111-71-7	98.0%	[50]
	Burnt	(E)-2-octenal	2548-87-0	98.0%	[51,52]

then conducted. Fig. 4 presents the concentration–response curves. As shown, all four sensors in the array exhibited clear and consistent responses to the selected substances. The response intensity increased with increasing analyte concentration, and distinct response patterns were observed for different odorants on the same sensor, indicating a certain degree of selectivity.

These results confirm that the proposed electronic nose system is capable of detecting VOCs generated during the cooking process in the oven and can capture dynamic changes in odor composition and concentration across different cooking stages. Notably, the sensor array demonstrated responsiveness to key marker compounds of the charring stage, which provides sensor-level evidence supporting the system's ability to detect burnt odors and discriminate among multiple cooking stages.

The GC–MS results also indicate that various odorant compounds are emitted at different stages of food roasting, which supports the cooking stage division. Through odor analysis, it is evident that the volatiles produced during cooking are highly diverse and complex. Identifying a single type of odor molecule that specifically signifies food maturity or charring presents a formidable challenge. Therefore, the use of broad-sensitive MOS gas sensors offers significant advantages. These sensors are capable of assessing the current state of the food from holistic odor levels, making them more suitable for detecting scenarios involving complex mixtures of substances. Coupled with machine learning algorithms, they can accurately classify the burnt state of food.

4.3. Water-cooling structure evaluation

Comparative experiments across the four recipes demonstrate that the integrated water-cooling structure significantly mitigates humidity within the sensor chamber.

The humidity distribution across all sampling windows was analyzed using box plots, as shown in Fig. 5. To isolate the humidity contribution of oven emissions from background, we calculated the differential humidity (ΔH). This was defined as the difference between the maximum humidity recorded within the sampling process and the minimum humidity during the preceding sweeping phase. Raw peak humidity values (without subtraction) are also provided in parentheses for reference. Outliers observed below the minimum whiskers in each recipe correspond to the initial roasting stages, where moisture release from the ingredients was negligible.

Our experimental results indicate that E-noses without a water-cooling system consistently displayed higher humidity values than those equipped with a water-cooling system. Without the water-cooling structure, the humidity for chicken wings was the highest, at 87.3% RH which is nearing the dew point humidity. In this study, chicken wings with a mass of 500 g were utilized. An increase in ingredient quantity or a substitution with ingredients could elevate the oven's humidity levels and potentially induce condensation within the E-nose chamber which may damage sensors and circuit boards. In contrast, the maximum humidity observed in the setup incorporating the water-cooling structure was only 35.0% RH (toast), suggesting that the water-cooling structure significantly reduces humidity levels and thereby ensures a

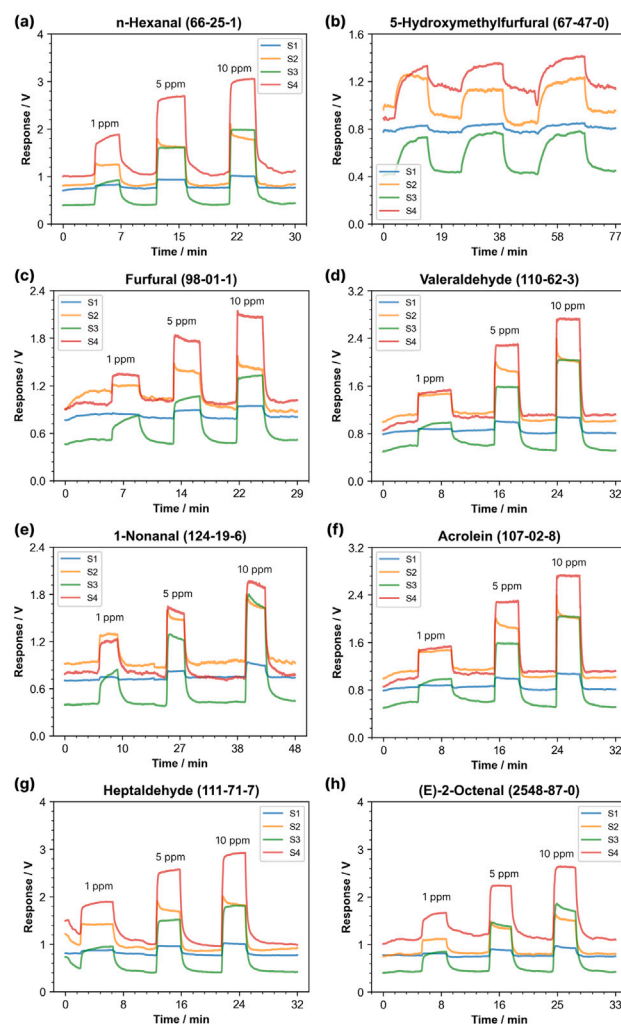


Fig. 4. Sensor responses to eight representative odorant compounds.

sufficient safety threshold for the continuous monitoring of oven gases by the E-nose.

Additionally, the water-cooling structure exhibited a marginal effect on the chamber temperature. The average temperature in both sensor chambers remained at approximately 45 °C, with a total rise of less than 5 °C during the entire process (detailed data in Supplementary Materials Section 7). This thermal stability suggests that the chamber's internal temperature is dominated by the self-heating of the MOS sensors rather than by the incoming oven gas. The sensors, housed in TO-packages, operate at approximately 350 °C, establishing a localized thermal equilibrium that largely overshadows external heat transfer. Moreover, the molecular activity of sampled gases at the

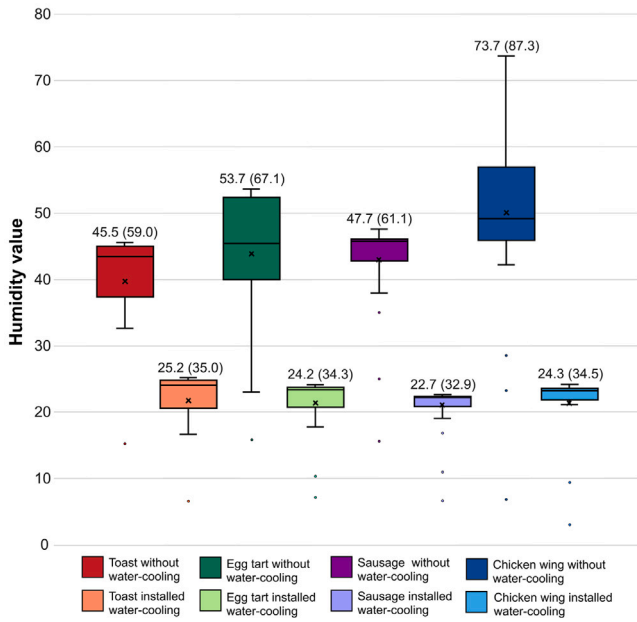


Fig. 5. Box plot of humidity values for four recipes. The original maximum humidity values are indicated within parentheses on the right side of the maximum values. The humidity value is retained to one decimal place.

sensor reaction sites presumably remains equivalent between the water-cooling and non-cooling chambers, preserving detection consistency and fidelity.

4.4. Classification of cooking stages

Feature extraction Feature extraction was applied to the raw electronic nose data, transforming multi-channel time-series data into high-dimensional feature vectors. Given the adoption of a window-based sampling strategy, the window served as the fundamental unit for feature extraction, modeling, and classification. The feature extraction method employed in this study comprises four sequential stages: calculating baseline values, capturing the ascending phase within a window, extracting features, and calibrating features. Baseline values were determined by averaging two data points prior to the beginning of each baking. In contrast to certain electronic nose systems that await signal stabilization, our system extracted features exclusively from the 4 s of sampling pump operation and the subsequent 2 s following the activation of sweeping pumps (covering the delay of gas paths). A 6-second data clip was defined as the ascending phases across all sensors. Maximum response values from four gas sensor signals, one temperature signal, and one humidity signal were extracted from these data windows, with baseline values subtracted to reduce environmental influences. These features were designated as $S1-S4$, T , H . Notably, each data window is not an isolated segment but rather an integral component of the entire baking process, which comprises tens of sequential windows. It is inherently linked to preceding odor evolution and data contexts. To capture this temporal dependency, four growth-rate features, denoted as $k1-k4$, were computed and appended to the feature vector. Furthermore, two temporal constraints were imposed based on the last model-predicted result: (1) the cooking state of food is irreversible (e.g., a state classified as burnt cannot revert to raw); and (2) the cooking state must evolve continuously without skipping intermediate stages (e.g., a direct transition from underdone to burnt is not permitted). The extraction formulas are shown as Formula (2a)–(2e), and Fig. 6(a) shows the extraction process.

$$FS_i^n = \text{Max}(S_i^n(t)) - \text{Baseline}(S_i), \quad (2a)$$

$$i = 1 \dots 4, n : \text{window_number}, 0 \leq t \leq 6s$$

$$FT^n = \text{Max}(T^n(t)) - \text{Baseline}(T), \quad (2b)$$

$$n : \text{window_number}, 0 \leq t \leq 6s$$

$$FH^n = \text{Max}(H^n(t)) - \text{Baseline}(H), \quad (2c)$$

$$n : \text{window_number}, 0 \leq t \leq 6s$$

$$k_i^n = FS_i^n - FS_i^{n-1} \quad i = 1 \dots 4, n \geq 1 \quad (2d)$$

$$k_i^0 = 0 \quad i = 1 \dots 4, n = 0$$

$$\text{if } \text{result} < \text{last_result} : \text{result} = \text{last_result} \quad (2e)$$

$$\text{if } \text{result} - \text{last_result} > 1 : \text{result} + = 1$$

$$0, 1, 2, 3 \text{ represents four stages}$$

Feature calibration Following the feature extraction of a given window, a calibration process was implemented to enhance data consistency. Although the food preparation followed the recipe instructions, the variability inherent in manual operations and differences in ingredient composition (e.g., water and protein content arising from the natural variability of raw food) could not be entirely eliminated. These factors resulted in variations in odor profiles and volatile compound concentrations, resulting in data discrepancies. Additionally, difference of environmental temperature and humidity, as well as potential aging effect in gas sensors were common causes of data inconsistency of electronic noses. These factors collectively led to fluctuations in the acquired data. To reduce the fluctuation and improve data consistency, a calibration process was applied, called *Translation Calibration*. We chose one time point for each recipe to translate the signals to a set level, 12 min for toast, 20 min for egg tarts and chicken wings, and 22 min for sausages respectively. The difference value, δ , between the sensor value at the selected time point and the set value was calculated. The subsequent newly acquired maximum values were shifted by a distance of δ , while keeping the data prior to the selected point unchanged. The time points were chosen at the middle of t_0 and t_1 , do not directly affect the recognition of cooking stages. Fig. 6(b) shows the translation process after the feature extraction. Fig. 6(c) shows the extracted odor vector input to the classification model.

Collected dataset In the first three E-noses (#1–#3), a total of 124 roasting trials were conducted. The specific number for each E-nose and recipe is detailed in Section 5 of Supplementary Materials. Utilizing a window-based sampling strategy with a 2-minute interval, raw sensor signals from each window were compressed into a single feature vector via feature extraction. This process yielded a comprehensive dataset of 2750 samples, comprising 477 for toast, 528 for egg tart, 808 for chicken wings, and 937 for sausages. To prevent data leakage arising from temporally adjacent windows of the same baking trial appearing in both the training and test sets, the dataset was partitioned at the trial level rather than the individual window level. Specifically, 80% of complete baking trials were randomly assigned to the training set and the remaining 20% to the test set for offline performance evaluation. The real-time test using E-nose #4 involved 20 additional roasting trials (averaging three per day) that were exclusively used for external verification and remained unseen during the model training.

Intra-device algorithm performance To identify the optimal classifier, we initially modeled the dataset using several machine learning algorithms, including Support Vector Machine (SVM), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Artificial Neural Networks (ANN). Comparative results (detailed in Supplementary Materials Section 8) indicate that XGBoost achieved superior classification accuracy. Further considering its light-weight tree-based architecture suitable for embedded systems, XGBoost was selected as the primary algorithm for the subsequent in-depth evaluation and edge-computing implementation. All hyperparameters were set to their default values. We first evaluated the classification performance based on intra-device dataset, which meant the training and testing odor data originated from the same electronic nose. After repeating data splitting, model training, hyperparameters tuning, and testing 20 times, the average test-set

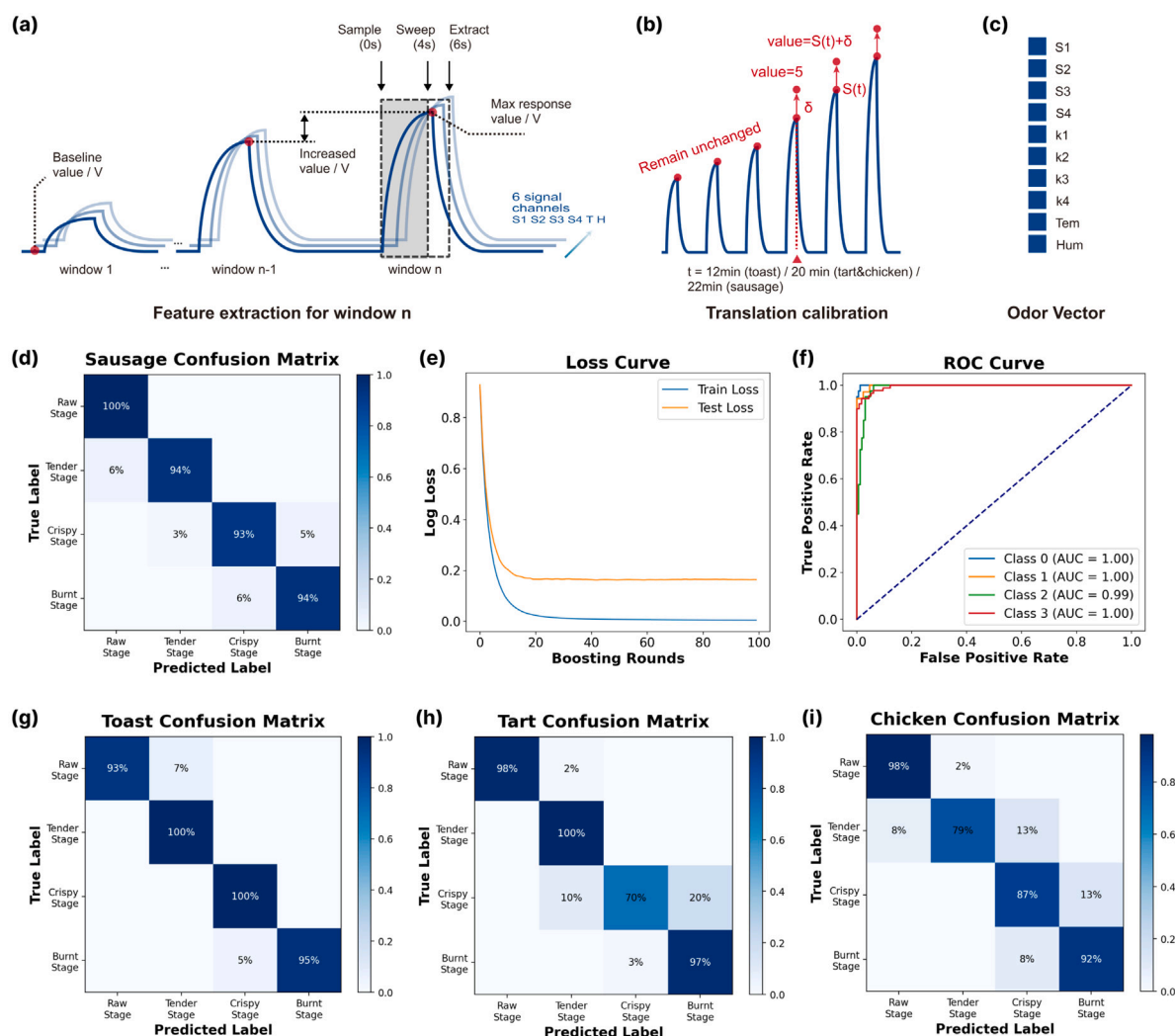


Fig. 6. Feature extraction process and XGBoost classification results. (a) Feature extraction for the data window n. Max responses, increased values compared with the last window (n-1), and the initial baselines were extracted respectively for all sensors (S1-S4,T,H) (b) Translation calibration. Features prior to the translation time point remained unchanged, while the data behind were translated by a calculated value δ . Only S1-S4 sensors were calibrated. (c) Extracted odor vector for each window. (d) Confusion Matrix of four stages classification, sausage using “all” dataset (Accuracy = 95.10%) (e) Loss curve of training and testing. (f) ROC curve and AUC values. (g) Confusion Matrix of toast classification (Accuracy = 95.83%) (h) Confusion Matrix of four tart classification (Accuracy = 95.45%) (i) Confusion Matrix of chicken classification (Accuracy = 91.06%).

accuracy and the confusion matrix of 4 cooking stages were obtained. Table 5 present the average test accuracy results of XGBoost. Results show that XGBoost model can reach an accuracy of classification over 90%, with the highest 93.72% in sausage recipe. Fig. 6(d)–(f) shows the confusion matrix for “All” dataset and sausage recipe, and the corresponding loss curve and ROC (Receiver Operating Characteristic) curve. Fig. 6(g)–(i) shows the confusion matrixes of toast, tart, and chicken respectively. The f1-score, recall, and precision performance are clearly shown in Supplementary Materials Section 9.

Cross-device algorithm performance In this section, we evaluated the model’s classification performance across different e-nose devices. Given the inherent inconsistency among gas sensors, training the model on one electronic nose and testing it on another provides a realistic measure of algorithmic robustness in practical applications. The training and testing protocols remained identical to the intra-device evaluation. We examined two reciprocal combinations: training on 95% of #1 dataset and testing on #2 dataset, and vice-versa. Table 5 (last two rows) summarize these results, where the system achieved an average accuracy of 88.94% and 89.73%. Notably, the high degree of symmetry between the reciprocal tests suggests that device-to-device variations were effectively minimized. This superior

cross-device consistency can be attributed to the standardized sensor integration and the stabilizing effect of the water-cooling structure on the sensing environment. These results suggest that the proposed E-nose system exhibits a degree of robustness against sensor non-uniformity, providing an encouraging basis for potential large-scale deployment in smart kitchen appliances.

Real-time on-board test To validate the feasibility of edge computing for intelligent kitchen appliances, the trained XGBoost algorithm was deployed directly onto the MCU for standalone inference. The model parameters were serialized into JSON format and ported to a lightweight Python library. To optimize memory consuming within the embedded environment, cross-compilation was employed to generate MicroPython bytecode (.mpy) files. This compression reduced the storage requirement for each recipe model.

The model was trained offline on a PC using datasets from E-noses #1-#3, while a new #4 was assembled for the real-time test. During operation, E-nose #4 performed the complete signal processing pipeline autonomously.

As summarized in Table 6, the average on-board inference latency was approximately 11.4 ms. The system achieved an average real-time accuracy of 91.3%. These on-board results show high consistency with

Table 5

Classification accuracy using XGBoost, intra-device and cross-device respectively.

E-nose	Toast/%	Egg tart/%	Chicken wing/%	Sausage/%	Average/%
#1 train&test	90.94 (± 2.74)	92.39 (± 3.39)	89.17 (± 2.65)	91.75 (± 3.02)	91.06
#2 train&test	92.19 (± 2.62)	93.52 (± 2.81)	85.83 (± 2.81)	89.60 (± 4.50)	90.29
#3 train&test	87.42 (± 1.42)	87.27 (± 3.96)	89.05 (± 4.62)	93.72 (± 5.51)	89.37
All(#1-#3) ^a	92.51 (± 1.90)	92.09 (± 1.99)	88.57 (± 2.36)	91.19 (± 1.27)	91.09
#1 train, #2 test	91.54 (± 1.26)	91.14 (± 1.64)	84.70 (± 0.72)	88.36 (± 2.75)	88.94
#2 train, #1 test	91.39 (± 0.61)	91.61 (± 0.49)	85.11 (± 2.88)	90.79 (± 1.27)	89.73

^a This row represents the three E-nose datasets were mixed together and evaluated.**Table 6**

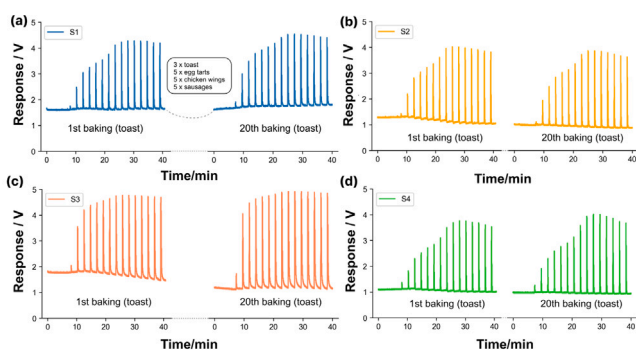
Results of on-board real-time test.

Performance	Toast	Egg tart	Chicken wing	Sausage	Average
On-board accuracy	90.67%	91.82%	91.72%	91.00%	91.30%
On-board inferring time	10.8 ms	11.0 ms	12.9 ms	10.9 ms	11.4 ms
Model size	42 KB	44 KB	61 KB	49 KB	49 KB

Table 7

Recipe generalizability performance.

Recipe	Accuracy	F1-score	Recall	Precision
One for all	89.43(± 1.72)	89.32(± 1.71)	89.43(± 1.72)	89.46(± 1.66)
Toast unseen	81.64(± 0.76)	82.35(± 0.69)	81.64(± 0.76)	85.26(± 0.65)
Egg tart unseen	86.34(± 0.71)	82.35(± 0.61)	86.34(± 0.71)	89.05(± 0.55)
Chicken unseen	78.95(± 0.96)	79.01(± 0.96)	78.95(± 0.96)	84.18(± 0.66)
Sausage unseen	82.14(± 0.60)	80.96(± 0.68)	82.14(± 0.60)	84.04(± 0.60)

**Fig. 7.** (a)–(d) Raw data of four gas sensors between the first and the 20th baking trials. No significant degradation in response range was observed, which demonstrated the sensor survived from the oven gas exposure.

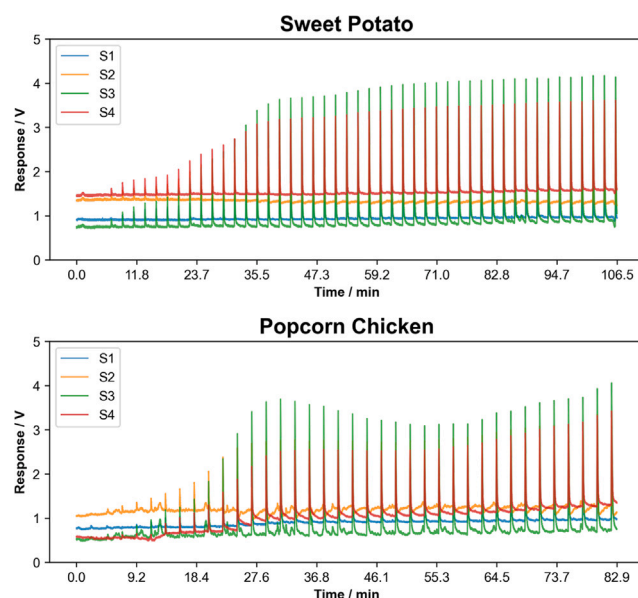
the offline PC-based evaluations, confirming the fidelity of the ported algorithm.

Crucially, the real-time campaign also served as a preliminary assessment of system stability under prolonged operation. Over the course of the validation, the system underwent 20 intensive roasting sessions (5 for each recipe), each persisting until the severe charring stage to maximize contaminant exposure. Fig. 7 contrasts the raw sensor response curves between the initial and final trials. Despite repeated exposure to high concentrations of oil mist and pyrolysis products, the MOS sensors exhibited comparable response amplitudes, and the baseline resistance drift remained within nominal limits (detailed analysis see Supplementary Materials Section 10). These observations suggest that the proposed water-cooling structure is effective in mitigating contamination of the sensing elements. However, it should be noted that the current evidence is based on a limited number of trials; a more systematic long-term stability study is warranted to fully characterize sensor degradation behavior.

4.5. Generalizability on novel recipes

In practical household scenarios, ovens are used for diverse recipes. Therefore, the ability of the E-nose system to generalize across different ingredients and handle unseen recipes is critical for system generalizability. To evaluate this capability, two training strategies were implemented: a unified model trained on all four recipes, and a “leave-one-recipe-out” validation to simulate the processing of novel recipes.

When the model was trained on a comprehensive dataset containing all four recipes (trial level, 80% training, 20% testing, 20 repeated

**Fig. 8.** (a) Signals for baking sweet potato. (b) Signals for popcorn chicken roasting.

shuffling and splitting), it achieved an accuracy of 89.43% comparable to single-recipe models as shown in Table 7. This indicates that, despite variations in VOC profiles across different ingredients, the system was able to extract common odor features associated with cooking stages. This is likely because severe carbonization breaks down organic matter into universal pyrolytic byproducts that overshadow recipe-specific aroma signatures, facilitating cross-recipe recognition of the burnt state.

In leave-one-recipe-out experiments, the model was trained on three recipes and tested on the held-out fourth. As shown in Table 7, the overall accuracy ranged from 78.95% to 86.34%, lower than the single-recipe model but still within an acceptable range. When chicken wing served as the unseen recipe, the training set consisted entirely of starch-based ingredients (toast and egg tart) and a processed meat product with mixed composition (sausage). As the only raw, whole-muscle ingredient in the dataset, chicken wing presents VOC release patterns least similar to any recipe in the training set, resulting in the lowest unseen-recipe accuracy.

To investigate the system’s broader applicability, odor profiles were collected from six additional recipes, including roasted sweet potatoes,

chiffon cake, egg cookies, steak, fish, and pork belly (see Fig. 8 and Supplementary Materials Section 11). Although these data were not included in the formal model due to single-trial limitations and the lack of sensory labels, they revealed consistent signal trends alongside the former recipes.

Analysis of the expanded recipes reveals three possible universal VOC evolution modalities, including asymptotic saturation, peak attenuation, and bimodal resurgence. The changing patterns reflect the common physicochemical transformations during oven cooking. These consistent kinetic trajectories provide a generalized feature set for cross-recipe stage recognition. The observations also suggest a hierarchical modeling strategy: a universal “coarse-grained” model could be effectively deployed for fundamental phase identification (e.g., raw, cooked, or burnt) by targeting signal turning points. In contrast, “fine-grained” optimization, such as distinguishing subtle tenderness or customized browning levels, would require recipe-specific datasets to account for unique volatile fingerprints. This dual-pathway approach demonstrates the potential of our E-nose system to balance general safety monitoring with specialized culinary precision in smart home ecosystems.

The generalizability of the E-nose to diverse cooking scenarios warrants further discussion. Through the unified model, leave-one-recipe-out validation, and trend observations on additional recipes, the current study has provided preliminary evidence that the system can extend beyond the specific recipes used during training. A notable practical advantage of this approach is that integrating a new recipe does not require large-scale data collection or model retraining — only a simple calibration reference point is needed, which can be estimated from cooking experience, reference recipes, or at most a single trial bake. This low-cost adaptability holds significant value for real-world product deployment, where supporting new recipes with minimal effort is essential.

Nevertheless, these findings should be interpreted with caution. The current validation is based on a limited number of ingredients under controlled experimental conditions, and more systematic analysis encompassing a broader range of food categories is required before definitive generalizability claims can be made. Moreover, real-world cooking introduces additional layers of complexity beyond recipe variety. Variations in ingredient quantities, diverse seasoning and preparation methods for the same recipe, and the highly personalized nature of individual cooking behaviors all give rise to VOC profile variations that may challenge the current model. Establishing robust generalizability across these dimensions will require more extensive investigation, potentially leveraging transfer learning frameworks to accommodate novel conditions with minimal recalibration.

5. Conclusion

This study presented an E-nose system for real-time cooking stage recognition and burnt odor detection, overcoming challenges in harsh gas adaptability and multi-recipe compatibility. By integrating a four-channel MOS sensor array with a water-cooling gas path, the system effectively mitigated the humidity and contamination in oven emissions. Through GC-MS analysis and static gas distribution tests, we identified distinctive VOC profiles across four cooking stages and validated the system's sensitivity to both cooking-related and burnt odors. Leveraging a window-based sampling strategy and the XGBoost algorithm, the system achieved a peak intra-device accuracy of 93.7% and maintained highest 89.7% accuracy in cross-device validation. On-board real-time tests demonstrated that the embedded microcontroller can execute standalone online inference without PC dependence, resulting in an average recognition accuracy of 91.3%. Furthermore, the unified model, “leave-one-recipe-out” validation, and data of six additional recipes provided preliminary evidence that the system can extend to unseen recipes, although further validation across a broader range of ingredients is needed to fully establish generalizability. Overall, this work provides a practical, low-cost, and high-accuracy solution for intelligent cooking monitoring, serving as a critical early-warning mechanism to enhance fire prevention and residential safety.

CRediT authorship contribution statement

Gang Yu: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuchi Sun:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haoxuan Han:** Writing – original draft, Methodology, Investigation. **Yang Lan:** Investigation. **Tao Guo:** Investigation. **Yulei Liu:** Methodology, Investigation. **Bangqiang Han:** Methodology, Investigation. **Xinjian Huang:** Writing – review & editing, Methodology, Conceptualization. **Wei Xi:** Resources, Funding acquisition. **Yan Cheng:** Resources, Funding acquisition. **Qi Lu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jia-Wei Yang:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Yingqing Xu:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.snb.2026.140161>.

Data availability

Data will be made available on request.

References

- [1] S. Hall, T. McGree, *Home cooking fires*, 2023.
- [2] X. Xu, P. Wang, N. Yu, H. Zhu, Experimental study on kitchen fire accidents in different scenarios, in: 2019 9th International Conference on Fire Science and Fire Protection Engineering, ICFSFPE, IEEE, 2019, pp. 1–6, <http://dx.doi.org/10.1016/j.jfp.2024.100338>.
- [3] W.-L. Hsu, J.-Y. Jhuang, C.-S. Huang, C.-K. Liang, Y.-C. Shiau, Application of internet of things in a kitchen fire prevention system, *Appl. Sci.* 9 (17) (2019) 3520, <http://dx.doi.org/10.3390/app9173520>.
- [4] E. Purlis, Browning development in bakery products – A review, *J. Food Eng.* 99 (3) (2010) 239–249, <http://dx.doi.org/10.1016/j.jfoodeng.2010.03.008>.
- [5] N. Tamanna, N. Mahmood, Food processing and maillard reaction products: Effect on human health and nutrition, *Int. J. Food Sci.* 2015 (2015) 1–6, <http://dx.doi.org/10.1155/2015/526762>.
- [6] K. Xiong, M. meng Li, Y. qiang Chen, Y. meng Hu, W. Jin, Formation and reduction of toxic compounds derived from the maillard reaction during the thermal processing of different food matrices, *J. Food Prot.* 87 (9) (2024) 100338, <http://dx.doi.org/10.1016/j.jfp.2024.100338>.
- [7] R. Tang, C. Pfrang, Indoor particulate matter (PM) from cooking in UK students' studio flats and associated intervention strategies: evaluation of cooking methods, PM concentrations and personal exposures using low-cost sensors, *Environ. Sci.: Atmospheres* 3 (3) (2023) 537–551, <http://dx.doi.org/10.1039/D2EA00171C>.
- [8] P. Das, D. Kemiseti, F.I. Jahan, S.E. Sriha, S.C. Raka, A comprehensive review on the formation of carcinogens from food products with respect to different cooking methods, *Br. J. Pharm. Res.* 33 (41B) (2021) 360–372, <http://dx.doi.org/10.9734/JPR1/2021/v33i41B32375>.
- [9] A. Dong, J. Lee, H.S. Shin, Influence of natural food ingredients on the formation of heterocyclic amines in fried beef patties and chicken breasts, *Food Sci. Biotechnol.* 20 (2) (2011) 359–365, <http://dx.doi.org/10.1007/s10068-011-0050-5>.

- [10] L. Zhang, F. Tian, H. Nie, L. Dang, G. Li, Q. Ye, C. Kadri, Classification of multiple indoor air contaminants by an electronic nose and a hybrid support vector machine, *Sens. Actuators B: Chem.* 174 (2012) 114–125, <http://dx.doi.org/10.1016/j.snb.2012.07.021>.
- [11] C. Wongchoosuk, C. Khunarak, M. Lutz, T. Kerdcharoen, WiFi electronic nose for indoor air monitoring, in: 2012 9th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology, IEEE, 2012, pp. 1–4, <http://dx.doi.org/10.1109/ECTICon.2012.6254166>.
- [12] J. He, L. Xu, P. Wang, Q. Wang, A high precise E-nose for daily indoor air quality monitoring in living environment, *Integration* 58 (2017) 286–294, <http://dx.doi.org/10.1016/j.vlsi.2016.12.010>.
- [13] B. Zong, S. Wu, Y. Yang, Q. Li, T. Tao, S. Mao, Smart gas sensors: recent developments and future prospective, *Nano Micro Lett.* 17 (1) (2025) 54, <http://dx.doi.org/10.1007/s40820-024-01543-w>.
- [14] K. Kumar, S.L. Tripathi, M. Mahmud, Internet of things enabled smart E-nose system for pollutants hazard detection and real-time monitoring in indoor mosquito repellents, *SN Comput. Sci.* 5 (5) (2024) 438, <http://dx.doi.org/10.1007/s42979-024-02786-5>.
- [15] A.J. Moshayedi, A. Sohail Khan, J. Hu, A. Nawaz, J. Zhu, E-nose-driven advancements in ammonia gas detection: a comprehensive review from traditional to cutting-edge systems in indoor to outdoor agriculture, *Sustainability* 15 (15) (2023) 11601, <http://dx.doi.org/10.3390/su151511601>.
- [16] O. Attallah, I. Morsi, An electronic nose for identifying multiple combustible/harmful gases and their concentration levels via artificial intelligence, *Measurement* 199 (2022) 111458, <http://dx.doi.org/10.1016/j.measurement.2022.111458>.
- [17] P. Khaloo, B. Oubre, J. Yang, T. Rahman, S.I. Lee, NOSE: a novel odor sensing engine for ambient monitoring of the frying cooking method in kitchen environments, *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3 (2) (2019) 1–25, <http://dx.doi.org/10.1145/3328920>.
- [18] F. Gerina, S.M. Massa, F. Moi, D. Reforgiato Recupero, D. Riboni, Recognition of cooking activities through air quality sensor data for supporting food journaling, *Hum. Centric Comput. Inf. Sci.* 10 (2020) 1–26, <http://dx.doi.org/10.1186/s13673-020-00235-9>.
- [19] Z. Haddi, J. Llano, M. Alfaras, D. Marín, A. Perera-Lluna, A. Fernández, X. Llauredó, N. Avellana, J. Fonollosa, E. Llobet, IoT gas sensors array for unobtrusive tracking of cooking activity, in: 2023 9th International Conference on Control, Decision and Information Technologies, CoDIT, IEEE, 2023, pp. 1–5, <http://dx.doi.org/10.1109/CoDIT58514.2023.10284485>.
- [20] Y.A. Sihombing, M.A. Ritonga, Detection of the cooking oil aroma by using a gas sensor in an electronic nose system, in: *Journal of Physics: Conference Series*, Vol. 1811, IOP Publishing, 2021, 012105, <http://dx.doi.org/10.1088/1742-6596/1811/1/012105>.
- [21] Y. Zhao, A. Li, R. Gao, P. Tao, J. Shen, Measurement of temperature, relative humidity and concentrations of CO, CO₂ and TVOC during cooking typical Chinese dishes, *Energy Build.* 69 (2014) 544–561, <http://dx.doi.org/10.1016/j.enbuild.2013.11.037>.
- [22] J. Pantelic, Y.J. Son, B. Staven, Q. Liu, Cooking emission control with IoT sensors and connected air quality interventions for smart and healthy homes: Evaluation of effectiveness and energy consumption, *Energy Build.* 286 (2023) 112932, <http://dx.doi.org/10.1016/j.enbuild.2023.112932>.
- [23] S.H. Hirano, J.R. Brubaker, D.J. Patterson, G.R. Hayes, Detecting cooking state with gas sensors during dry cooking, in: *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 2013, pp. 411–414, <http://dx.doi.org/10.1145/2493432.2493523>.
- [24] F.S. Fedorov, A. Yaqin, D.V. Krasnikov, V.A. Kondrashov, G. Ovchinnikov, Y. Kostyukovich, S. Osipenko, A.G. Nasibulin, Detecting cooking state of grilled chicken by electronic nose and computer vision techniques, *Food Chem.* 345 (2021) 128747, <http://dx.doi.org/10.1016/j.foodchem.2020.128747>.
- [25] B. di Diodoro, C. Bax, N. Dellarosa, F. Corazza, G. Langfelder, L. Capelli, Bread baking monitoring by smart sensory system: a feasibility study, in: 2022 IEEE International Symposium on Olfaction and Electronic Nose, ISOEN, IEEE, 2022, pp. 1–4, <http://dx.doi.org/10.1109/ISOEN54820.2022.9789664>.
- [26] C. Bax, B. di Diodoro, A. Ticozzi, N. Dellarosa, F. Corazza, G. Langfelder, L. Capelli, Smart odour sensing for automated monitoring of bread products, in: *Proceedings*, Vol. 97, MDPI, 2024, p. 172, <http://dx.doi.org/10.3390/proceedings2024097172>.
- [27] B.J. Lotesoriere, L. Viareggi, C. Bax, L. Capelli, Real-time monitoring of bread baking in ovens by smart odour sensors: focus on calibration transfer, in: 2024 IEEE International Symposium on Olfaction and Electronic Nose, ISOEN, IEEE, 2024, pp. 1–3, <http://dx.doi.org/10.1109/ISOEN61239.2024.10555994>.
- [28] H. Zhang, X. Wang, X. Shen, X. Li, B. Wu, G. Li, H. Bai, X. Cao, X. Hao, Q. Zhou, Z. Yao, Chemical characterization of volatile organic compounds (VOCs) emitted from multiple cooking cuisines and purification efficiency assessments, *J. Environ. Sci.* 130 (2023) 163–173, <http://dx.doi.org/10.1016/j.jes.2022.08.008>.
- [29] C. Sun, L. Zhao, X. Chen, L. Nie, A. Shi, H. Bai, G. Li, A comprehensive study of volatile organic compounds from the actual emission of Chinese cooking, *Environ. Sci. Pollut. Res.* 29 (35) (2022) 53821–53830, <http://dx.doi.org/10.1007/s11356-022-19342-4>.
- [30] H. Liu, J. Li, D. Zhang, N. Hamid, D. Liu, W. Hua, C. Du, Q. Ma, H. Gong, The effect of thermal times of circulating non-fried roast technique on the formation of (non)volatile compounds in roasted mutton by multi-chromatography techniques and heat transfer analysis, *Food Res. Int.* 174 (2023) 113567, <http://dx.doi.org/10.1016/j.foodres.2023.113567>.
- [31] S. Wang, H. Liu, T. Xie, N. Zhang, J. Sun, H. Chen, B. Sun, Study on volatile aroma compounds in donkey broths of different stewing time, *Flavour Fragr. J.* (2) (2022) 37, <http://dx.doi.org/10.1002/ffj.3689>.
- [32] M. Rivai, D. Aulia, Use of electronic nose to identify levels of cooking cookies, *IEEE Access* (2024) <http://dx.doi.org/10.1109/ACCESS.2024.3428322>.
- [33] H. Zhang, Y. Liu, Q. Li, X. Zhao, M. Sun, J. Hou, C. Guo, C. Huang, X. Cheng, L. Huo, et al., Oxygen vacancy dual regulation strategy realizes wide humidity range monitoring of UIO-66 (ce)-derived CeO₂ for intelligent food manufacturing, *J. Mater. Chem. A* (2025) <http://dx.doi.org/10.1039/D5TA02082D>.
- [34] P. French, G. Krijnen, F. Roozeboom, Precision in harsh environments, *Microsyst. Nanoeng.* 2 (1) (2016) 1–12, <http://dx.doi.org/10.1038/micronano.2016.48>.
- [35] J. Hao, W. Lu, D. Yin, W. Hao, Y. Wang, A humidity-tolerant NO₂ sensor based on Ag₃PO₄-SnSe₂ heterostructures, *New J. Chem.* 47 (7) (2023) 3242–3248, <http://dx.doi.org/10.1039/D2NJ06076K>.
- [36] Y. Wang, C. Tang, M. Su, Y. Ji, L. Xie, Q. Yang, A. Du, Y. Zhou, J. Yang, Highly sensitive, humidity-tolerant and flexible NO₂ sensors based on nanoplate Bi₂Se₃ film, *Chin. Chem. Lett.* 34 (8) (2023) 107981, <http://dx.doi.org/10.1016/j.cclet.2022.107981>.
- [37] S. Mor, B. Gunay, M. Zanotti, M. Galvani, S. Pagliara, L. Sangaletti, Current opportunities and trends in the gas sensor market: A focus on e-noses and their applications in food industry, *Chemosensors* 13 (5) (2025) 181, <http://dx.doi.org/10.3390/chemosensors13050181>.
- [38] N.K. Mahanti, S. Shivashankar, K.B. Chhetri, A. Kumar, B.B. Rao, J. Aravind, D. Swami, Enhancing food authentication through E-nose and E-tongue technologies: Current trends and future directions, *Trends Food Sci. Technol.* (2024) 104574, <http://dx.doi.org/10.1016/j.tifs.2024.104574>.
- [39] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, <http://dx.doi.org/10.48550/arXiv.1603.02754>.
- [40] Y. Sun, G. Yu, Q. Lu, H. Han, J.-W. Yang, Y. Xu, An electronic nose device with rapid and universal odor detection capability, *Sens. Actuators B: Chem.* (2025) 137871, <http://dx.doi.org/10.1016/j.snb.2025.137871>.
- [41] M. Maire, B. Rega, M.-E. Cuvelier, P. Soto, P. Giampaoli, Lipid oxidation in baked products: Impact of formula and process on the generation of volatile compounds, *Food Chem.* 141 (4) (2013) 3510–3518, <http://dx.doi.org/10.1016/j.foodchem.2013.06.039>.
- [42] J. Pico, I. Khomenko, V. Capozzi, L. Navarini, F. Biasioli, Real-time monitoring of volatile compounds losses in the oven during baking and toasting of gluten-free bread doughs: A PTR-MS evidence, *Foods* 9 (10) (2020) 1498, <http://dx.doi.org/10.3390/foods9101498>.
- [43] S. Makhoul, A. Romano, V. Capozzi, G. Spano, E. Aprea, L. Cappellin, E. Benozzi, M. Scampicchio, T.D. Märk, F. Gasperi, et al., Volatile compound production during the bread-making process: Effect of flour, yeast and their interaction, *Food Bioprocess Technol.* 8 (9) (2015) 1925–1937, <http://dx.doi.org/10.1007/s11947-015-1549-1>.
- [44] G. Giovannelli, C. Cappa, 5-hydroxymethylfurfural formation in bread as a function of heat treatment intensity: Correlations with browning indices, *Foods* 10 (2) (2021) 417, <http://dx.doi.org/10.3390/foods10020417>.
- [45] A.C. Lemos, V.S. de Borba, P.T. Scaglioni, E. Badiale-Furlong, Processing parameters in breadmaking and bioaccessibility of acrylamide and 5-hydroxymethylfurfural, *Food Res. Int.* 174 (2023) 113523, <http://dx.doi.org/10.1016/j.foodres.2023.113523>.
- [46] F.R. Chaoyue Feng, Effect of heat-treatment on volatile flavor compounds in Egg Yolk, *Meat Res.* 20 (10) (2006) 31–33, <http://dx.doi.org/10.7506/rljy1001-8123-200610009>.
- [47] H. Jiang, M. Zhang, J. Ye, M. Qian, X. Li, W. Zhao, W. Bai, HS-SPME-GC-MS and OAV analyses of characteristic volatile flavour compounds in salt-baked drumstick, *Lwt* 170 (2022) 114041, <http://dx.doi.org/10.1016/j.lwt.2022.114041>.
- [48] Y. Shu, F. Wu, W. Yang, W. Qi, R. Li, Z. Zhang, Dietary effect of curcumin on amino acid, fatty acid, and volatile compound profiles of chicken meat, *Foods* 13 (14) (2024) 2230, <http://dx.doi.org/10.3390/foods13142230>.
- [49] K. Jiang, C. Huang, F. Liu, J. Zheng, J. Ou, D. Zhao, S. Ou, Origin and fate of acrolein in foods, *Foods* 11 (13) (2022) 1976, <http://dx.doi.org/10.3390/foods11131976>.
- [50] M. Zang, L. Wang, Z. Zhang, K. Zhang, D. Li, X. Li, S. Wang, H. Chen, Changes in flavour compound profiles of precooked pork after reheating (warmed-over flavour) using gas chromatography-olfactometry-mass spectrometry with chromatographic feature extraction, *Int. J. Food Sci. Technol.* 55 (3) (2020) 978–987, <http://dx.doi.org/10.1111/ijfs.14306>.
- [51] Q. Zhang, W. Qin, D. Lin, Q. Shen, A.S. Saleh, The changes in the volatile aldehydes formed during the deep-fat frying process, *J. Food Sci. Technol.* 52 (12) (2015) 7683–7696, <http://dx.doi.org/10.1007/s13197-015-1923-z>.

- [52] J. Bi, Z. Lin, Y. Li, F. Chen, S. Liu, C. Li, Effects of different cooking methods on volatile flavor compounds of chicken breast, *J. Food Biochem.* 45 (8) (2021) e13770, <http://dx.doi.org/10.1111/jfbc.13770>.

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